Ma Yuexiao

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Education

Ph.D. in Intelligent Science and Technology September 2022 – Present
Department of Artificial Intelligence, Xiamen University, Xiamen, China
Advisor: Prof. Rongrong Ji
Co-Advisor: Assoc. Prof. Xiawu Zheng

M.S. in Intelligent Science and Technology September 2020 – June 2022
Department of Computer Science, Xiamen University, Xiamen, China
Advisor: Prof. Rongrong Ji
Co-Advisor: Assoc. Prof. Xiawu Zheng

B.S. in Mathematics and Applied Mathematics September 2016 – June 2020
Northeastern University, Shenyang, China

Research Experience

ByteDance Data | Beijing
Research Intern, Quantization Team Jun 2022 – Oct 2025

- **Seedance acceleration**: Accelerated Seedance generation models for low-cost, low-latency inference. Proposed **SplitQuant**, a low-bit quantization scheme that enables lossless W4A4 image generation and supports fused SplitQuant kernels, achieving up to **3×** **end-to-end speedup** for single Linear operators across different input shapes.
- **LLM quantization**: Co-developed **AffineQuant** (ICLR'24), a post-training quantization framework that applies affine transformations to reduce activation outliers and quantization error. Took part in method design and experiments on Llama-2, reducing quantization error by **32%**.
- **Vision Transformer quantization**: Designed the outlier-aware slicing mechanism for ViT post-training quantization (ICML'24). The project targeted efficient INT4 deployment of vision backbones; my contribution improved ImageNet accuracy by **7%** under INT4 quantization.

Co-author: Xuefeng Xiao

Publication

Key: † = Corresponding Author, * = Equal Contribution

Flow Caching for Autoregressive Video Generation (ICLR 2026)
Yuexiao Ma*, Xuzhe Zheng*, Jing Xu*, Xiwei Xu, Feng Ling, Xiawu Zheng, Huafeng Kuang, Huixia Li, Xing Wang, Xuefeng Xiao, Fei Chao, Rongrong Ji†

Outlier-aware Slicing for Post-Training Quantization in Vision Transformer (ICML 2024)
Yuexiao Ma, Huixia Li, Xiawu Zheng, Feng Ling, Xuefeng Xiao, Rui Wang, Shilei Wen, Fei Chao, Rongrong Ji†

AffineQuant: Affine Transformation Quantization for Large Language Models (ICLR 2024, score: 8 8 8 5)
Yuexiao Ma, Huixia Li, Xiawu Zheng, Feng Ling, Xuefeng Xiao, Rui Wang, Shilei Wen, Fei Chao, Rongrong Ji†

Solving Oscillation Problem in Post-Training Quantization Through a Theoretical Perspective (CVPR 2023)
Yuexiao Ma, Huixia Li, Xiawu Zheng, Xuefeng Xiao, Rui Wang, Shilei Wen, Xin Pan, Fei Chao, Rongrong Ji†

OMPQ: Orthogonal Mixed Precision Quantization (AAAI 2023)
Yuexiao Ma, Taisong Jin†, Xiawu Zheng, Yan Wang, Huixia Li, Yongjian Wu, Yunsheng Wu, Guannan Jiang, Wei Zhang, Rongrong Ji

Positive solutions of fourth-order problems with dependence on all derivatives in nonlinearity under Stieltjes integral boundary conditions (Boundary Value Problems, 2019)
Yuexiao Ma, Chenyang Yin, Guowei Zhang†

An Information Theory-inspired Strategy for Automatic Network Pruning (IJCV 2025)

Xiawu Zheng*, **Yuexiao Ma***, Teng Xi, Gang Zhang, Errui Ding, Yuchao Li, Jie Chen, Yonghong Tian, Rongrong Ji†

Motion-Aware Caching for Efficient Autoregressive Video Generation (ICML 2026)

Jing Xu*, **Yuexiao Ma***, Xuzhe Zheng, Xing Wang, Shiwei Liu, Chenqian Yan, Xiawu Zheng, Rongrong Ji, Fei Chao, Songwei Liu†

Training-Free Multimodal Large Language Model Orchestration (ICML 2026)

Tianyu Xie, **Yuexiao Ma**, Yuhang Wu, Wang Chen, Jiayi Ji, Tat-Seng Chua, Xiawu Zheng†, Rongrong Ji

A^{2}RBench: An Automatic Paradigm for Formally Verifiable Abstract Reasoning Benchmark Generation (ICML 2026)

Qingchuan Ma, **Yuexiao Ma**, Yongkang Xie, Tianyu Xie, Xiawu Zheng†, Rongrong Ji

Polybasic Speculative Decoding Through a Theoretical Perspective (ICML 2025)

Ruilin Wang, Huixia Li, **Yuexiao Ma**, Xiawu Zheng, Fei Chao, Xuefeng Xiao, Rongrong Ji†

Automated Fine-Grained Mixture-of-Experts Quantization (ACL Findings, 2025)

Zhanhao Xie, **Yuexiao Ma**, Xiawu Zheng, Fei Chao, Wanchen Sui, Shen Li, Yong Li, Rongrong Ji†

Dynamic Post-Training Quantization for Diffusion Models (IJCNN, 2025)

Xiao Yang, Huixia Li, lijiang Li, Xiawu Zheng, **Yuexiao Ma**, Feng Ling, Jie Wu, Xuefeng Xiao, Rui Wang, Shilei Wen, Fei Chao†

♡ Professional Service

- **Journal Reviewer**: IEEE Transactions on Image Processing (TIP), IEEE Transactions on Multimedia (TMM)
- **Conference Reviewer**:
  - *Reviewer*: ECCV'24, NeurIPS '25 (**Top Reviewer**), ICML '25, ICLR '25, AAAI'26, ICLR'26 (**Top Reviewer**), CVPR'26, ICML'26 (**Top Reviewer**)